

Product Requirements Document (PRD)

PRD : **Campus Connect –AI Driven Community and Study Co-ordination**

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Executive Summary

Problem Statement

University students struggle to manage their academic and social coordination outside official Learning Management Systems. Information about study groups, peers, and academic collaboration is scattered across multiple noisy platforms, making it difficult to find the right people to study with at the right time. This fragmentation leads to missed learning opportunities, increased academic stress, and social isolation.

Market Opportunity

Existing tools like LMS platforms and messaging apps (WhatsApp, Telegram, Discord) support communication but lack intelligent, AI-driven study group matching tailored to academic needs.

Product Overview

Campus Connect is a unified, AI-powered platform that organizes students' academic schedules, social interactions, and collaborative learning in one place. It eliminates scattered tools by providing intelligent matching into balanced, relevant study groups based on courses, skills, schedules, and personalities, reducing app-switching and mismatched teams.

Primary Business Goals

- Boost student engagement by 30% through higher daily active users and session times.
- Cut academic stress by 25% via better organization and support.
- Increase collaborative learning participation by 20% compared to ad-hoc methods.

Target Users and Persona

Primary Users:

University students (undergraduate/graduates, ages 17-24) seeking better organization for academic and social interactions.

Persona: Anjali, the First-Year Undergraduate

"I joined 15 groups but don't know who to study with—everyone's busy or flakes"



Anjali, The First-Year Undergraduate

NAME: ANJALI

AGE: 18 GENDER: FEMALE LOCATION: DELHI

JOB: B.TECH COMPUTER SCIENCE STUDENT TRAIT: BUDGET CONSCIOUS

BACKGROUND

DAILY ROUTINE & TECH HABITS:

- Wakes at 8 AM; classes 9-5 PM; studies 2-3 hours, nightly + weekends.
- App usage: 4+ hours /day on WhatsApp (100+ groups), Telegram (class channels), Instagram (stories/DMs); switches apps 50+ times daily but ignores 70% notifications due to overload.
- Tools: Phone screen cluttered with 20+ group chats, Google Calendar half-synced, no central hub.

GOALS AND MOTIVATIONS:

- Excel academically (GPA >8.0) to secure internships at FAANG-like firms.
- Build a core friend group (5-10 people) for studies and campus events within the first semester.

BEHAVIOURS:

- Prefers quick matches (under 2-3 mins), rates of peers on reliability post-session.
- Would pay ₹99/semester for perfect matches.

PAIN POINTS:

- Scattered info: Misses 2-3 study sessions/month because events are buried in WhatsApp forwards.
- Group formation: Spends 30-60 min/week DMing strangers; ends up with "free riders" who don't contribute.
- Social friction: Hesitant to cold-message seniors, feels excluded from "cool" groups.

Demographics

- Age: 18, female, first-year B.Tech Computer Science student at a mid-tier Indian university (e.g., in Delhi).
- Lives in a hostel, from a tier-2 city, budget-conscious.

Daily Routine & Tech Habits

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Quotes

- "I joined 15 groups but don't know who to study with—everyone's busy or flakes."

Behaviors

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Primary Needs for Campus Connect

- AI matches in <1 min based on course (CS101), schedule (evenings free), skills (strong Python, weak DSA), and vibe (quiz: introvert, reliable).
- One-tap join with icebreakers; post-match ratings to refine future groups.

Solution Overview (MVP)

MVP Feature: AI-Driven Study Group Matchmaker

For the initial MVP, Campus Connect will use a **Wizard of Oz approach** to simulate AI-driven matching while minimizing engineering complexity.

Users see automated flows, but you manually create matches using a spreadsheet.

The goal is to validate user demand, matching logic, and engagement before building a fully automated AI system.

User Flow:

Stage	Action	Backend Reality (The "Wizard")
1. Entry	Student verifies university email via SSO.	Validate domain against whitelist (e.g., @iitd.ac.in, @du.ac.in) using simple regex in Google Forms/Zapier. Log user ID/timestamp.
2. Profiling	Student selects: • Course/subject • Confidence (Beginner/Intermediate/Advanced) • Time slots • Goals (revision/doubt-solving/exam prep).	Data auto-pipes to Airtable/Google Sheet via Zapier/Typeform. New row: UserID, email, profile JSON, timestamp.
3. Matching	User sees "AI finding your best peers..." (spinner 30-60s).	Open Sheet, scan last 10-20 profiles, manually group 3-5 by: same subject + balanced confidence + overlapping slots. Note group ID in Sheet.
4. Review	User sees match: "Your group: Anjali (Advanced, Mon 7PM), Raj (Intermediate, exam prep)" with Accept/Decline + "Regenerate?"	You copy group details from Sheet → paste into Zapier/email template → trigger personalized notification within 5 min.
5. Success	3+ accepts → "Group confirmed!" + one-tap join chat.	Auto-create group in free Telegram bot/Discord → share invite link. Or manual: Add all to pre-made WhatsApp group named "CS101 Study - [GroupID]".
6. Feedback	Post-session survey: "How balanced was the group? 1-5"	Responses pipe back to Sheet. You review weekly to refine matching rules (e.g., "avoid 3 beginners together").

Out of Scope:

- Eventdiscovery, clubs, mentoring, full social feed.
- Real AI (add post-validation).

Non-Goals: Campus Connect is not intended to replace LMS platforms or become a full social network. The MVP deliberately avoids events, clubs, and feeds to focus exclusively on validating study group matching.

User Stories and Acceptance Criteria (MVP)

User Story 1: Profile Input

As Anjali, **I want to** enter my course and study preferences **so that I** can be matched with relevant peers.

Acceptance Criteria:

- **Given** Anjali opens the profile form, **when** she selects a course from the dropdown and ≥ 1 time slot, **then** the submit button enables and form validates successfully.
- **Given** Anjali leaves course/time fields empty, **when** she tries to submit, **then** validation errors appear and submission is blocked.
- **Given** Anjali completes the form correctly, **when** she submits, **then** she sees "Profile saved! finding matches" (<2s) and data saves to backend.

User Story 2: Balanced Matching

As Anjali, **I want to** be matched with students at a similar pace **so that** study sessions are effective.

Acceptance Criteria:

- **Given** ≥ 3 compatible profiles exist in the pool, **when** matching runs, **then** Anjali receives a group of 3–5 with $\geq 80\%$ course/time overlap and balanced confidence levels.
- **Given** insufficient compatible profiles exist, **when** matching attempts to run, **then** Anjali sees "Building your group!" message.
- **Given** a match is generated, **when** the wizard approves it, **then** Anjali receives notification within 10 minutes .

User Story 3: Match Preview

As Anjali, **I want to** view group details before joining **so that I** feel comfortable accepting the match.

Acceptance Criteria (Given-When-Then):

- **Given** a match is ready, **when** Anjali opens the preview screen, **then** she sees group size, subject, goal, member names/confidence levels, and Accept/Decline buttons.
- **Given** Anjali clicks Accept, **when** ≥ 3 members accept, **then** chat opens immediately with success confirmation.
- **Given** Anjali clicks Decline, **when** she selects reason, **then** system logs feedback and offers "Regenerate match?" option.

User Story 4: Group Chat

As Anjali, **I want** a group chat **so that** I can stay updated.

Acceptance Criteria:

- **Given** ≥ 3 members accept the match, **when** chat opens, **then** Anjali sees auto-generated icebreaker message and all group members listed.
- **Given** Anjali sends a message, **when** another member is online, **then** they see real-time delivery status and typing indicators.
- **Given** < 3 members are active, **when** Anjali tries to chat, **then** chat shows read-only mode with "Waiting for more members" message.

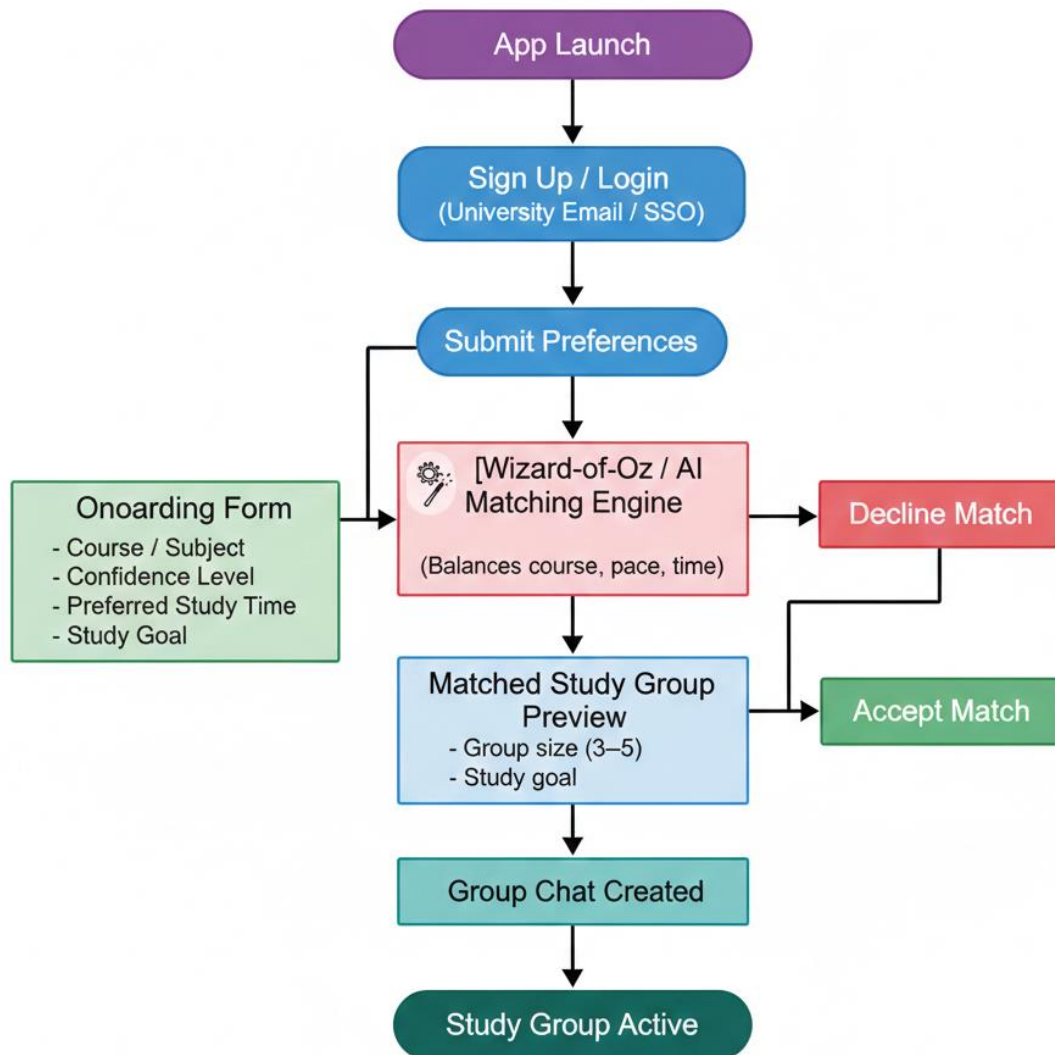
User Story 5: Flexible Exit

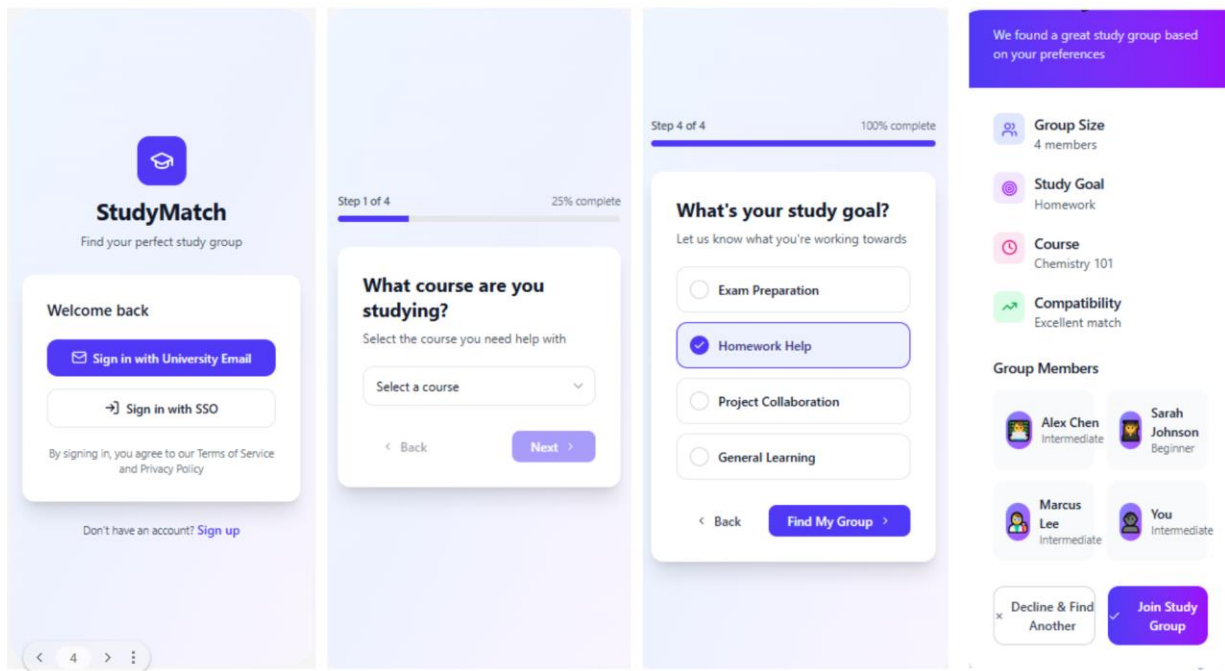
As Anjali, **I want to** leave a group if it's not helpful **so that** I'm not stuck.

Acceptance Criteria :

- **Given** Anjali is in a group, **when** she clicks "Leave Group", **then** confirmation modal appears explaining impact on other members.
- **Given** Anjali confirms leave, **when** group size updates, **then** remaining members receive notification and updated member count.
- **Given** Anjali is the final member to leave, **when** group dissolves, **then** survey prompt appears asking "Why did this group fail?"

User Flow Diagram





Technical Requirements

Now (MVP)

This section outlines the minimum technical setup required to validate problem–solution fit using a Wizard-of-Oz MVP.

1. Performance Requirements

Requirement	Specification	Test Criteria
Profile Form Submit	<3s response	Manual timer test
Match Notification	<15min delivery	Email logs
Daily Capacity	50 new users / 10 groups	Sheet row count

2. Scalability & Availability

- Max Users: 200 students (1 hostel/freshers batch)
- Uptime: 99% (4h downtime/month acceptable)
- Your Manual Effort: <30min/day matching

3. Security & Privacy

- Auth: University email verification (manual domain validation during MVP)
- Data: Google Sheet (private sharing: you + 1 tester)

- Chat: Public WhatsApp/Telegram groups (manual creation)

4. Data Storage

- Single Google Sheet "CampusConnect_MVP"
- Columns:
 - A: UserID (auto),
 - B: Email,
 - C: Course,
 - D: Confidence,
 - E: TimeSlots (CSV),
 - F: Goals (CSV),
 - G: MatchGroupID,
 - H: Status (pending/accepted/left),
 - I: Timestamp

5. Integrations (MVP - No Code)

Service	Purpose	Setup
Typeform	Profile intake	Free account + public link
Zapier	Form → Sheet	Free tier (100 tasks/mo)
Gmail	Match notifications	Gmail template
WhatsApp/Telegram	Group chat	Manual group creation + share links

6. Tech Stack

- Frontend: Typeform (hosted form)
- Backend: Google Sheets + Zapier workflows
- Notifications: Gmail templates
- Chat: WhatsApp/Telegram groups (manual)
- Dashboard: Your personal Sheet view

Total Cost: ₹0 (free tiers)

Setup Time: 1 hour

Next(Post WOz Validation)

Phase 2: Real MVP (After 100 Users, Week 4)

Add:

- Custom React landing page (Vercel free)
- Firebase Auth (SSO)
- Rule-based matching (Apps Script)
- Automated chat bot (group creation)

Cost: ₹500/month

Scale: 100 users

Later (Post MVP Validation)

Phase 3: Full Product (After 1000 Users, Month 3)

1. React Native app
2. PostgreSQL + Node.js backend
3. ML matching (cosine similarity)
4. Push notifications (FCM)
5. Google Calendar sync

Cost: ₹5000/month

Scale: 10k users

Migration Trigger: WOz hits 60% acceptance → auto-graduate to Phase 2.

Launch Success Metrics (KPIs):

The success of the Campus Connect MVP will be measured using a small set of quantifiable metrics aligned with the AARRR funnel, focusing specifically on validating the AI-driven (Wizard-of-Oz) study group matching feature.

1. Acquisition

Metric: Sign-Up Conversion Rate

- Definition: Percentage of invited students who complete sign-up using a university email
- Formula: $(\# \text{ of completed sign-ups} / \# \text{ of students invited}) \times 100$
- Success Target: $\geq 30\%$

Validates whether the problem resonates strongly enough for students to try the product.

2. Activation

Metric: Match Activation Rate

- Definition: Percentage of signed-up users who complete their profile and receive a study group match
- Formula: $(\# \text{ of users receiving a match} / \# \text{ of signed-up users}) \times 100$
- Success Target: $\geq 80\%$

Measures how effectively users reach the core value moment: being matched to a study group.

3. Activation (Core Value Validation)

Metric: Match Acceptance Rate

- Definition: Percentage of users who accept a suggested study group
- Formula: $(\# \text{ of accepted matches} / \# \text{ of matches shown}) \times 100$
- Success Target: $\geq 60\%$

Directly validates the perceived quality and relevance of the matching logic.

4. Retention

Metric: First Study Session Participation Rate

- Definition: Percentage of matched users who participate in at least one study session within 7 days
- Formula: $(\# \text{ of users attending first session} / \# \text{ of users who accepted a match}) \times 100$
- Success Target: $\geq 70\%$

Confirms that matches lead to real-world academic collaboration, not just sign-ups.

KPI Summary

Funnel Stage	Metric	Target
Acquisition	Sign-Up Conversion Rate	$\geq 30\%$
Activation	Match Activation Rate	$\geq 80\%$
Activation	Match Acceptance Rate	$\geq 60\%$
Retention	First Session Participation	$\geq 70\%$

MVP Success Criteria

Success = Kill/Continue Decision

Outcome	Criteria	Action
WIN	$\geq 3/4$ KPIs hit targets	Build real AI MVP (Phase 2)
MAYBE	$2/4$ KPIs hit	Tweak matching rules + 2nd cohort
KILL	$\leq 1/4$ KPIs hit	Pivot or abandon

Graduation Trigger: 60%+ acceptance rate $\rightarrow$ invest ₹25k in Phase 2 dev.

Risk and Mitigation

Below Risks must be considered through-out the MVP phase.

Risk	Impact	Likelihood	Mitigation
Low match acceptance	MVP fails to validate value	Medium	Iterate matching criteria weekly based on feedback
Manual matching overload	Slows response time	Medium	Cap daily users; batch matches once per day
Incomplete user profiles	Poor-quality matches	High	Make key fields mandatory during onboarding
Drop-off after match	Weak retention signal	Medium	Send reminder nudges before first study session
Privacy concerns	User distrust	Low	Limit access to sheets; use university emails only

Appendix

Term	Definition
MVP (Minimum Viable Product)	The smallest version of the product built to validate problem–solution fits with real users.
Wizard of Oz MVP	An MVP approach where complex functionality (e.g., AI matching) is simulated manually while appearing automated to users.
AI-Driven Matching	The process of recommending study groups based on user inputs such as course, availability, and learning pace.
AARRR Funnel	A product growth framework covering Acquisition, Activation, Retention, Referral, and Revenue.
Activation	The moment a user experiences the core value of the product (receiving a study group match).
Match Acceptance Rate	Percentage of users who accept a suggested study group.
Time-to-Match	Time taken from profile submission to receiving a study group recommendation.
Study Group Matchmaker	Core MVP features that form small, compatible study groups.
Problem–Solution Fit	Evidence that a product effectively solves a real user problem.
SSO (Single Sign-On)	Authentication method allowing users to log in using an institutional email system.